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SEAT No. _____

No. of Printed Pages : 2

Sardar Patel University
External Examination (ATKT)
M.Sc. (Information Technology) Semester – I
PS01CINT05: Operating System Concepts
20th April, 2017 Thursday

Time: 10:00 AM to 01:00 PM

Max. Marks: 70

Q-1 Choose the most appropriate option for each question.

[08]

- (i) ___ Operating system guarantees that critical task completes on time.
 - a) Soft real time b) Hard real time
 - c) Hard Time sharing d) Soft time sharing
- (ii) Full form of SMP is _____.
 - a) Symmetric Multiprocessing b) Symmetric Microprocessing
 - c) Synchronous Multiprocessing d) Synchronous Microprocessing
- (iii) The mechanism that brings a page into memory only when it is needed is called _____.
 - a) Segmentation b) Fragmentation
 - c) Demand Paging d) Page Replacement
- (iv) ___ Model maps many user level threads to one kernel thread.
 - a) Many to One b) One to Many
 - c) Many to Many d) All of these
- (v) ___ provide an interface to the services made available by an operating system.
 - a) Function call b) System call
 - c) API d) All of these
- (vi) The control is under operating system when the system is in ___ mode.
 - a) User b) Observer
 - c) Kernel d) Guest
- (vii) Which of the following feature provides reliability in RAID?
 - a) Cloud storage b) Stripping
 - c) Data dictionary d) Redundancy
- (viii) Which of the following is not a type of the file?
 - a) Text b) Object
 - c) Audio d) None of these

Q-2 Answer ANY SEVEN from the following

[14]

- (i) What is process? List all states of process.
- (ii) Differentiate between shared memory system and message passing system.
- (iii) Explain Virtual memory in short.
- (iv) Differentiate between: Preemptive & Non Preemptive Scheduling.
- (v) Define: Seek Time and Rotational Latency
- (vi) List any four attributes of the file.
- (vii) List benefits of multithreading.
- (viii) Differentiate between Windows family operating systems and Unix.
- (ix) Draw the structure of PCB.

Q-3

A. Explain functions performed by an operating system. [06]

B. Explain monolithic structure of OS. [06]

OR

B. Explain necessary condition to solve critical section problem.

Q-4

A. What is Paging? Explain basic method of paging. [06]

B. Explain deadlock with suitable example. Also explain necessary conditions which characterize deadlock. [06]

OR

B. Explain contiguous memory allocation scheme. What do you mean by fragmentation?

Q-5

A. What do you mean by disk scheduling algorithm? List various disk scheduling algorithms; explain any one of them in detail. [06]

B. What is RAID? List various levels of RAID. Explain any two layers in brief. [06]

OR

B. Write a note on magnetic disk as a mass storage structure.

Q-6

A. What is file? Explain different access methods for the file. [06]

B. Write a note on directory structure. [06]

OR

B. Explain following Unix commands: (i) ls (ii) mv (iii) rm

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